

Money Sums

Time limit: 1.00 s

Memory limit: 512 MB

You have n coins with certain values. Your task is to find all money sums you can create using these coins.

Input

The first input line has an integer n : the number of coins.

The next line has n integers $x_1, x_2, \dots, x_n$: the values of the coins.

Output

First print an integer k : the number of distinct money sums. After this, print all possible sums in increasing order.

Constraints

- $1 \leq n \leq 100$
- $1 \leq x_i \leq 1000$

Example

Input	Output
4 4 2 5 2	9 2 4 5 6 7 8 9 11 13